What follows is an itemized list on how a developer resource may configure their Integrated Development Environment(IDE).
NOTE: This list of instructions has been published for the **Eclipse LUNA IDE**.

	Instructions	Description / Comments
1	Request Elevated Rights on Developer Laptop	Submit Service NOW Ticket to request ADMIN Rights to your LOCAL MACHINE. Follow the steps below: Click on the following Service NOW Link **Description:** Local Desktop Administration - Add User Fill out the following information: **Business Unit**: Allstate Technology and Strategic Ventures **Division**: Infrastructure Services & Innovation **Department**: Infrastructure Applications **Location**: South **Machine Name**: Either found at "MyComputer" or found by typing "HOSTNAME" at the command prompt **Business Need**: New developer resource with Allstate Benefits Effort, <*entertEffortNameHere*>, here in Jacksonville, Florida. Require either FULL Administrative Rights or Elevated Rights, to LOCAL Machine, in order to install a DEVELOPER LOAD on such machine. Such is required to effectively support the V3 Application, as well as provide necessary bug fixes, patches, and or enhancements upon business partner(s) request.
2	Download **Node.js**	Allstate Approved Versions, **8.11.3 +**, can be found at the following link: https://nodejs.org/en/ I chose to download the Windows 64 bit Version as an MSI File. ******It is critical that Step 1 was accomplished first in order to run this MSI File******
3	Right-Click on the **Node.js** MSI File and "**Run Elevated with DefendPoint**"	This will walk through a typical windows install process. The only nuance that I chose to update the install path to "**C:\nodejs**" from the default. "**C:\Program Files\nodejs**". Run this to successful completion.
4	Create a shortcut on your desktop for the "**Node.js**" Command Prompt	Found beneath folder structure: **C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Node.js**
5	Update the "**PATH**" System Environment Variable for Node.js	This is to compensate and validate that the value is pointing at the destination folder set in **Step #3.**
6	Create the following System Environment Variables for Node.js Installs: a. NODE_TLS_REJECT_UNAUTHORIZED = 0 b. GIT_SSL_NO_VERIFY = 1	This is a pre-requisite. In order to perform **Node.js Package Installs using the "npm" a Node Package Manager**, at the **Command Prompt, Node.js** EXPECTS a Signed Certificate from a qualified Certificate Authority (CA). Unfortunately when first installing packages from the "**Node.js**" **Command Prompt,** the developer may not have such certificates LOCAL to their machine. These System Environment Variables provide the necessary means around that. Also, this is a common practice when a developer needs to install and configure, **Node.js** behind a Corporate Web Proxy. **For further information, reference:** **a.** How to Setup Behind a Corporate Proxy: https://jjasonclark.com/how-to-setup-node-behind-web-proxy/ **b.** SSL Certificates Explained: https://medium.com/nodejs-tips/ssl-certificate-explained-fc86f8aa43d4

	Instructions	Description / Comments
7	Common Practice to restart your computer here...	Restart Computer Now
8	Right-Click on the Desktop Shortcut **Node.js** and "**Run Elevated with DefendPoint**" ***It is a always good practice to run applications as "**Run Elevated with DefendPoint**"	Opens **Node.js Command Prompt**
9	To Continue the setup from **behind a "Corporate Web Proxy"** Update the "**Node.js Configuration File.**	Navigate to to where **Node.js** was installed. For me it was the following folder structure: "**C:\nodejs**" Type the following statements at the "**npm**" Command Line and hit the ENTER Key to update the "**Node.js**" Configuration File:. **a.** npm config set strict-ssl=false **b.** npm config set proxy=http://webproxy.igslb.allstate.com:8080/ **c.** npm config set https-proxy=http://webproxy.igslb.allstate.com:8080/
10	Make one additional update to the "**Node.js npm**" Configuration File.	Type the following statements at the "**npm**" Command Line and hit the ENTER Key. -- This will point **Node.js** at the **Allstate Artifactory Location** **a.** npm config set registry=https://artifactory.allstate.com:443/artifactory/api/npm/npm-cf/
11	Install **Allstate APPFabric CLI & Package Dependencies**	Type the following statements at the "**Node.js**" Command Line and hit the ENTER Key. **a.** npm login This will log the developer into the **Artifactory** that was just configured as the **REGISTRY** in the previous **Step #10** Once successfully logged in, Type the following statements at the "Node.js" Command Line and hit the ENTER Key. **b.** npm install -g app-fabric-cli **T**his will install the **APPFabric** Framework to your local machine
12	Create System Environment Variables for **LOCAL_USER** & **LOCAL_PASSWORD**	Both System Environment Variables are the credentials necessary to access the **Artifactory**, mentioned in the previous step. Without them you will be unable to build projects. This will become important around **Step #37**, when we install the application build software "**Gradle**" and start the process of building our application.

	Instructions	Description / Comments
		The values will be dependent upon the **Developers' Network Credentials** and **NOT** the "**System User ID & Password** " that was established for the project through **MarketPlace**. Reference the following for building an application: Team Confluence Page: Import an Existing Project & Build Team Confluence Page: Create a New Project & Build
13	Validate that the "**Node.js** " Configuration File has been updated.	Type the following statements at the "**npm**" Command Line and hit the ENTER Key. **a.** npm config edit *****A Copy has been attached for the reader's reference***** ? Unknown Attachment
14	Install **Node.js Express Packages & Dependencies**	Type the following statements at the "**Node.js**" Command Line and hit the ENTER Key. **a.** npm install By default, npm install will install all modules listed as dependencies in package.json. Found beneath folder structure: ../node_modules/npm. This will automatically install all dependencies. If successful, there will be a series of Command Prompt Console Outputs written to the screen. Screen shot captured below for reference.

	Instructions	Description / Comments

Select Administrator: c:\windows\system32\cmd.exe

```
glob@7.1.1 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\glob
fs.realpath@1.0.0 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\fs.realpath
inflight@1.0.6 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\inflight
once@1.4.0 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\once
wrappy@1.0.2 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\wrappy
inherits@2.0.3 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\inherits
minimatch@3.0.3 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\minimatch
brace-expansion@1.1.11 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\brace-expansion
balanced-match@1.0.0 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\balanced-match
concat-map@0.0.1 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\concat-map
path-is-absolute@1.0.1 C:\nodejs\node_modules\npm\node_modules\tacks\node_modules\path-is-absolute
up to date in 2.739s

17 packages are looking for funding
  run `npm fund` for details

npm notice created a lockfile as package-lock.json. You should commit this file.
added 399 packages from 349 contributors in 67.893s

17 packages are looking for funding
  run `npm fund` for details

C:\nodejs\node_modules\npm>
```

If the "npm" Command resulted in failure, install dependencies manually by executing following commands:

"npm install express -g"
"npm install url -g"
"npm install fresh -g"
"npm install cookie -g"
"npm install methods -g"
"npm install crc -g"
"npm install send -g"
"npm install connect -g"
"npm install commander -g"

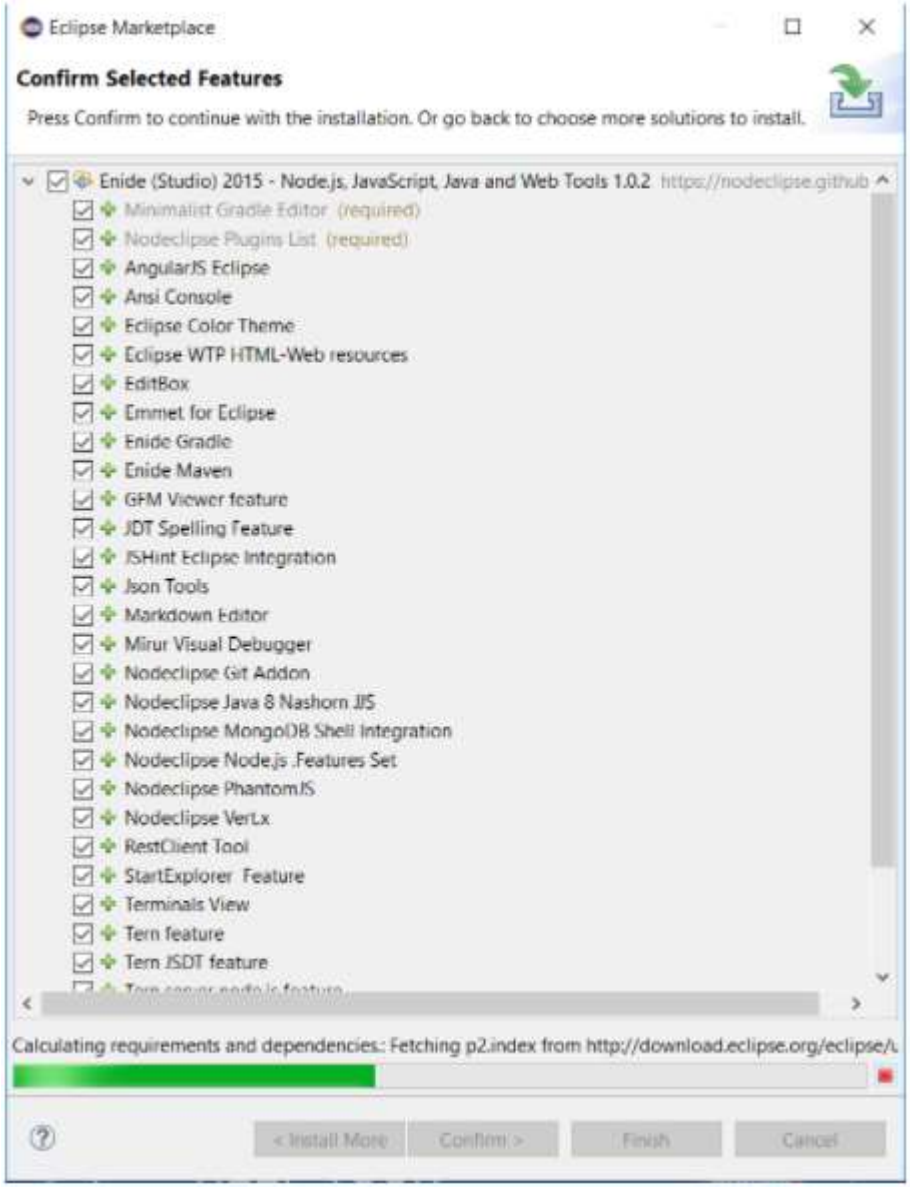
*****For additional Reference*****

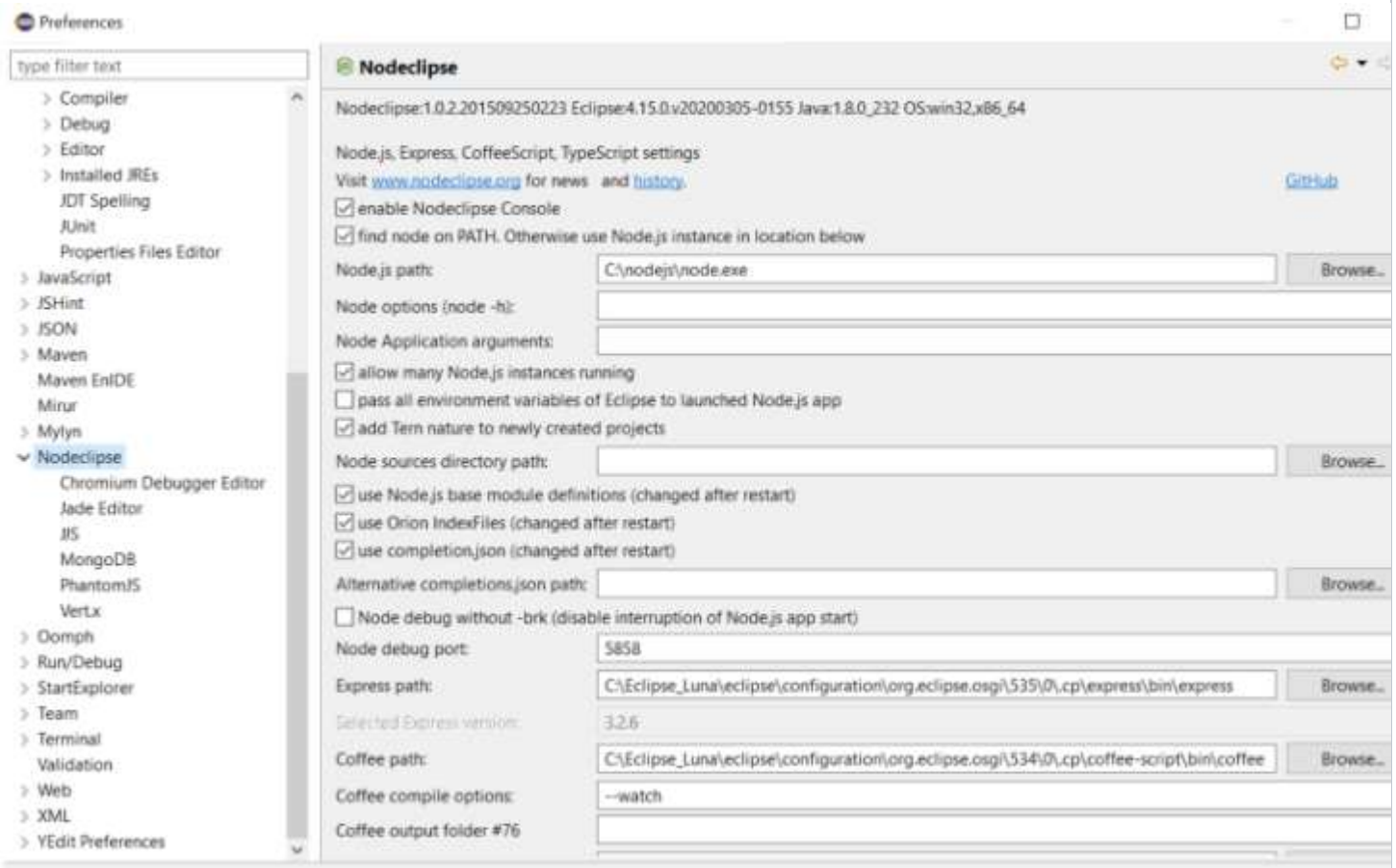
https://coderwall.com/p/mbov6w/running-nodejs-and-express-on-windows

	Instructions	Description / Comments
15	Install **Nodeclipse CLI, Express, & Express Generator Package Dependencies**	Type the following global statements at the "**Node.js**" Command Line and hit the ENTER Key. **a.** npm install -g nodeclipse **b.** npm install -g express **c.** npm install -g express-generator These are dependencies for the Eclipse IDE Nodeclipse Plugin mentioned later.
16	**OPTIONAL:** Additional Package Installs specific to Microsoft .NET and Visual Studios IDE's npm install --global windows-build-tools	Installing the "**Windows-build-tools**" Packages Globally, will allow the developer to install Microsoft Windows Build Tools specific to the .NET World. It will also install Microsoft Visual C++ Build Tools that caters towards the Microsoft Visual Studios IDE,. All being required to Compile Native Node.js Modules. It will also install **Python Ver 2.7,** configuring the developer machine and "**npm**" accordingly to allow the developer to leverage "**node-gyp**". "**Node.js**" **Build Utilities** This is not a requirement for this effort, because we will *not alter/create/update Node.JS Modules nor are we developing within a Client-side Framework, like React JS. Also all development will be performed in JAVA 8, not Node.js.*
17	Download **JAVA**	Allstate Approved Version JAVA Version **8.0 +**, can be found at Oracle's site; https://www.oracle.com/java/technologies/javase-jdk8-downloads.html I chose to download the Windows 64 bit Version as an EXE File, "**jdk-8u251-windows-x64.exe**". ******It is critical that Step 1 was accomplished first in order to run this EXE File****** JAVA Platform Standard Edition 8 Documentation: https://docs.oracle.com/javase/8/docs/
18	Right-Click on the **jdk-8u251-windows-x64.exe** Executable File and "**Run Elevated with DefendPoint**"	This will walk through a typical windows install process. The only nuance that I chose to update the install path to "**C:\Java**" from the default. "**C:\Program Files (x86)\Java\jre1.8.0_251**". Run this to successful completion.
19	Create/Update the JAVA_HOME System Environment Variable	Validate/Configure the JAVA_HOME System Environment Variable Path has been set to the destination folder where the Java was installed to. Currently 1.8.251 is installed on my machine. Presently the JAVA_HOME System Environment Variable has been configured with an "AdoptOpenJDK" Declaration. May not have to update the System Environment Variable after all. Will require further research as to how this compares to the Oracle Version just downloaded. At this point perform any level of cleanup that may have System Environment Variables pointing at previously leveraged JDK/JRE's... This is the current value for the JAVA_HOME System Environment Variable: "**C:\Program Files\AdoptOpenJDK\jdk-8.0.232.09-hotspot**"
20	Download Eclipse Luna Integrated Development Environment (IDE)	As before, download the 64 bit version; https://www.eclipse.org/luna/

	Instructions	Description / Comments
21	Right-Click on the **eclipse-inst-win64.exe** Executable File and "**Run Elevated with DefendPoint**"	When the Eclipse Installer Window appears select the following: "**Eclispe IDE for JAVA Developers**" The following installation settings were selected: **a. Java 1.8+ VM:** "C:\Program Files\AdoptOpenJDK\jre-8.0.232.09-hotspot(current)" **b. Installation Folder: "C:\Eclipse_Luna**" from the default. "**C:\Users\mjojg\eclipse\java-2020-03**". Run this to successful completion. **c.** Accept the License Agreement Once complete it will offer the following options: a. Launch b. Show README File c. Open in System Explorer d. Keep Installer I chose to Launch the IDE to continue with the configuration process.
22	Validate that the Eclipse IDE, JAVA Compiler and Installed JRE's are configured to **Step #'s 5 & 6**	Proceed to the following IDE Navigation: **Windows → Preferences → JAVA → Compiler** **Windows → Preferences → JAVA → Installed JRE's**
23	Install Nodeclipse Eclipse Plugin NOTE: Reference: the following link if necessary https://www.dev2qa.com/node-js-hello-world-example-with-eclipse/	**Navigate to the following:** **Help → Eclipse Market** Type "**Nodeclipse**" within the **Find Search Box** **Hit the ENTER Key** The following will be returned, screenshot provided below. It will redirect the developer to a "**Nodeclipse & Enide**" Website; https://nodeclipse.github.io/

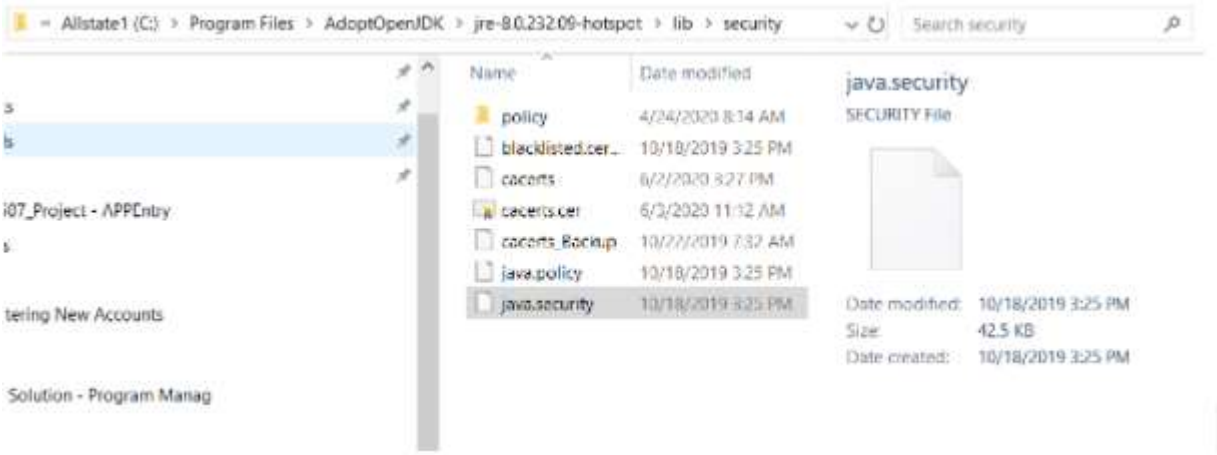
	Instructions	Description / Comments
2 4	Drag and Drop the "**Nodeclipse & Enide**" Eclipse Plugin into MarketPlace;	Eclipse will start the process of plugin install. Select all features as shown below and continue.

	Instructions	Description / Comments
		
25	Agree to the Install License	Agree to the Eclipse Plugin Install License and Continue

	Instructions	Description / Comments
26	Restart Eclipse	Restart Eclipse.
27	Validate that Nodeclipse points at the newly installed version of "**Node.js**" Destination Folder (***Reference Step #3***)	Confirm Installed "**Node.js**" Destination Folder by performing the following navigation; **Window → Preferences → Nodeclipse** Reference Screen Shot Below. 
28	Install Spring Tools Eclipse Plugin	**Navigate to the following:** **Help → Eclipse Market** Type "**Spring Tools**" within the **Find Search Box** **Hit the ENTER Key**

	Instructions	Description / Comments
		Click the "**Install**" Button. (Reference the Screen-shot below for "*Spring Tools 4 (aka Spring Tool Suite 4) 4.6.2 Release*") **NOTE:** Eclipse will start the process of plugin install. Select all features as shown below and continue.
29	Agree to the Install License	Agree to the Eclipse Plugin Install License and Continue
30	Restart Eclipse	Restart Eclipse.
31	Install GitHub. **NOTE:** There are two options to install GIT. One is a Command Line Utility that I have personally chosen to download; whereas the other is a Desktop version of the application. If a previous version exists on your system, please uninstall before proceeding.	**GitHub Command Line Utility:** https://git-scm.com/book/en/v2/Getting-Started-Installing-Git **Installing on Windows** There are also a few ways to install Git on Windows. The most official build is available for download on the Git website. Just go to https://git-scm.com/download/win and the download will start automatically. Note that this is a project called Git for Windows, which is separate from Git itself; for more information on it, go to https://gitforwindows.org. Download the Windows 64 bit Version as an EXE File.

	Instructions	Description / Comments
		It is critical that Step 1 was accomplished first in order to run this EXE File ---------- **GitHub Desktop APP** can be found at the following link: https://desktop.github.com/ Download the Windows 64 bit Version as an EXE File. ***It is critical that Step 1 was accomplished first in order to run this EXE File***
32	Right-Click on the **Git-2.27.0-64-bit.exe** File and "**Run Elevated with DefendPoint**"	This will walk through a typical windows install process. If a previous version exists on your system, UNINSTALL it before proceeding. I have chosen to install to the destination folder, **C:\Git**, instead of the default. "**C:\Program Files\Git**". Run this to successful completion. Once successfully completed you may Click the "Next" Button It will redirect you to the Release Notes: file:///C:/Git/ReleaseNotes.html
33	Navigate to the following : Start → Programs → Git.	Right Click on **Git Bash and Open File Location**
34	Create Shortcut on Desktop for GitBash Command Prompt	File Explorer Window will Open. Right-Click on "Git Bash" and navigate to; **Send To → Desktop(Create Shortcut**
35	Copy the KeyStore Certificate **cacerts** File out to your local JRE	Reference the following Keystore Certificate Project; https://github.allstate.com/Compozed/java-bp-cacerts From this project, simply copy the **cacerts file** to your systems JAVA JRE system folder path. It has also been provided here for reference. Make certain to backup the previous one prior to copying out the new certificate file. Reference the "cacerts_Backup" as an example (Mine has been provided as an example and a screen shot follows); C:\Program Files\AdoptOpenJDK\jre-8.0.232.09-hotspot\lib\security

	Instructions	Description / Comments
		 Allstate1 (C:) › Program Files › AdoptOpenJDK › jre-8.0.232.09-hotspot › lib › security Search security Name Date modified policy blacklisted.cer cacerts cacerts.cer cacerts_Backup java.policy java.security java.security SECURITY File Date modified: 10/18/2019 3:25 PM Size: 42.5 KB Date created: 10/18/2019 3:25 PM Solution - Program Manag ? Unknown Attachment
36	Prepare to configure **Git** andor **GitHub for a Corporate Web Proxy**	Based upon the project mentioned in **Step #32**, https://github.allstate.com/Compozed/java-bp-cacerts, the Keystore Certificate File, **cacerts** was then leveraged as a base to generate a Bundled certificate file "**ca-bundle.crt**" This Bundled Certificate File is what will be used to configure the Git and GitHub Web Proxy Settings. Both Bundled Files have been provided below. ***For your reference what follows are the steps to create the Bundled Certificate File.*** ***Within the "certmgr" utility at the command line.*** It provides the means of **importing** the "cacerts" keystore file, only to then **export** it as a **Base64 Encoded Format** and **append it to the "ca-bundle.crt" File.** Also, from Google Chrome navigate to Allstate's GitHub Website, **https://github.allstate.com/** and **export** the **certificate** only to **repeat the process mentioned previously to append it to the "ca-bundle.crt" File.** ***Within Google Chrome:*** ***a. Click on the menu (:) displayed on the top right corner after the browser address bar*** ***b. Now follow More tools >> Developer Tools.*** ***c. Select the Security tab, a second right option with default settings. Click on View Certificate and go to the "Details" you will have certificate details*** ? Unknown Attachment ? Unknown Attachment **For the Git Command Line Utility copy both bundles out to the following system folder path (*Remember this is specific to my machine and where I installed the Git Command Line Utility*);** C:\Git\usr\ssl\certs

	Instructions	Description / Comments
		For the GitHub Desktop Application copy both bundles out to the following system folder path (*Remember this is specific to my machine, my user, mjojg, you will have to alter this to your user path*); C:\Users\mjojg\AppData\Local\GitHubDesktop\app-2.5.2\resources\app\git\usr\ssl\certs
37	Right-Click on the newly created Desktop Shortcut **GitBash** and "**Run Elevated with DefendPoint**" ***It is a always good practice to run applications as "**Run Elevated with DefendPoint**"	Should open a GitBash Terminal
38	Configure Git for your profile	Let's make sure that you have git installed and perform the initial configuration. Don't forget to substitute my name, email, and project with your values. Reference, https://chronicler.tech/windows-password-manager-and-git/, for further information. Execute the following GitBash Statements. Hit the ENTER Key after each statement **a.** git --version **b.** git version 2.27.0.windows.1 **c.** git config --global user.email ***insertEmailAddressHere*** **d.** git config --global user.name "***insertFirstLastNameHere***"
39	Configure Git for the Corporate Web Proxy	Execute the following GitBash Statements. Hit the ENTER Key after each statement. **If using the Git Command Line Utility** Perform **Steps a & b,** this will allow you to enter your credentials upon using Git. For further information please reference the following; https://chronicler.tech/windows-password-manager-and-git/ Please note this will be specific to your user, so please replace '***DOMAIN\\mjojg***' with your username. **a.** git config --global http.proxy http://DOMAIN\\mjojg@github.allstate.com **b.** git config --global https.proxy http://DOMAIN\\mjojg@github.allstate.com **If using the GitHub Desktop Application** Perform **Steps c & d,** this will clear the proxy settings **c.** git config --global http.proxy "" **d.** git config --global https.proxy "" Perform the remaining **Steps e & f** **e.** git config --global http.sslverify false **f.** git config --global http.sslCAInfo ***C:/Git/usr/ssl/certs/ca-bundle.crt***

	Instructions	Description / Comments
		I chose to use the GitHub Desktop Application, so when finished your ***.gitconfig file***, located at your User System Folder path, mine has been provide for reference; ***C:\Users\mjojg,*** should resemble the following(Attached below); ? Unknown Attachment
40	Install Gradle	Reference the following link for downloads: https://gradle.org/install/ **Microsoft Windows users** Create a new directory C:\Gradle with File Explorer. Open a second File Explorer window and go to the directory where the Gradle distribution was downloaded. Double-click the ZIP archive to expose the content. Drag the content folder gradle-6.5 to your newly created C:\Gradle folder.
41	Update System Environment Variable "PATH"	Update the System Environment Variable "PATH" with the following "C:\Gradle\gradle-6.5\bin"
42	Validate successful installation	Right-Click on the Command Prompt Desktop Shortcut and "**Run Elevated with DefendPoint**" Execute the following statement: **a.** gradle -v If successful, should return the following: -- Picked up _JAVA_OPTIONS: -Xmx6144m Welcome to Gradle 6.5! Here are the highlights of this release: - Experimental file-system watching - Improved version ordering - New samples For more details see https://docs.gradle.org/6.5/release-notes.html -- Gradle 6.5 -- Build time: 2020-06-02 20:46:21 UTC Revision: a27f41e4ae5e8a41ab9b19f8dd6d86d7b384dad4

	Instructions	Description / Comments
		Kotlin: 1.3.72 Groovy: 2.5.11 Ant: Apache Ant(TM) version 1.10.7 compiled on September 1 2019 JVM: 1.8.0_232 (25.232-b09) OS: Windows 10 10.0 amd64
43	Close out all applications and restart your computer	
	Congrats!!! You are now ready to Create a New Project or Import an Existing One.	